Current Flow in a Good Conductor

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Lecture Objective

By the end of this lecture, you will be prepared to:

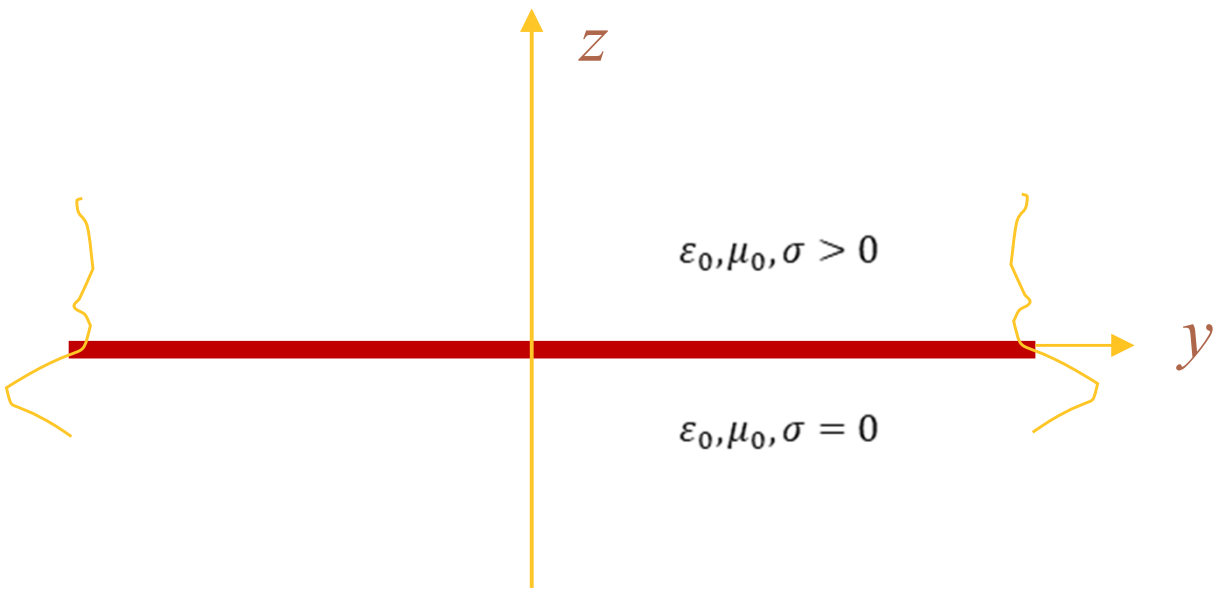
- Describe how current behaves in a good conductor.
- Explain the concept of surface impedance.

Current Flow in a Good Conductor

- At DC, the current flow uniformly through the cross section of a wire.
- In the time-varying case, the skin effect causes the current to be maximum at the surface of the conductor and decrease exponentially as a function of distance towards the axis of the wire.

Semi-Infinite Conducting Material

- Consider the following scenario:
- For $z<0$, we have free space
- For $z>0$, we have a non-magnetic conducting medium



Semi-Infinite Conducting Material

- Assume a plane wave traveling in the +z-direction is incident on the boundary at $z = 0$.
- The electric field just below the boundary is given by

$$\tilde{\vec{E}}(z = 0^-) = \hat{x}E_0$$

- The BCs mandate that the electric field be continuous across the boundary.

Semi-Infinite Conducting Material

- The electric and magnetic fields at any depth in the conductor are given by

$$\tilde{\vec{E}}(z) = \hat{x}E_0 e^{-\alpha z} e^{-j\beta z}$$
$$\tilde{\vec{H}}(z) = \hat{y}\frac{E_0}{\eta_c} e^{-\alpha z} e^{-j\beta z}$$

Semi-Infinite Conducting Material

- For a good conductor, we have

$$\alpha = \beta = \sqrt{\pi f \mu_0 \sigma}$$

$$\delta_s = \frac{1}{\sqrt{\pi f \mu_0 \sigma}}$$

$$\eta_c = (1 + j)\sqrt{\frac{\pi f \mu_0}{\sigma}}$$

Semi-Infinite Conducting Material

- The current in the conducting medium is given by

$$\tilde{\vec{J}}(z) = \sigma \tilde{\vec{E}}(z) = \hat{x}\sigma E_0 e^{-\alpha z} e^{-j\beta z}$$

$$= \hat{x}\sigma E_0 e^{-\frac{z}{\delta_s}} e^{-j\frac{z}{\delta_s}}$$

Semi-Infinite Conducting Material

- Consider the total current flowing in the x-direction through a strip of width w in the y-direction and infinite depth in the z-direction.

$$\tilde{I} = \int_0^{\infty} \int_0^{w} \tilde{\vec{J}}(z) \cdot \hat{x} dy dz = \hat{x} \sigma E_0 w \int_0^{\infty} e^{-(1+j)\frac{z}{\delta_s}} dz$$

$$= \left(\frac{\sigma E_0}{1+j} \right) w \delta_s$$

Semi-Infinite Conducting Material Continued

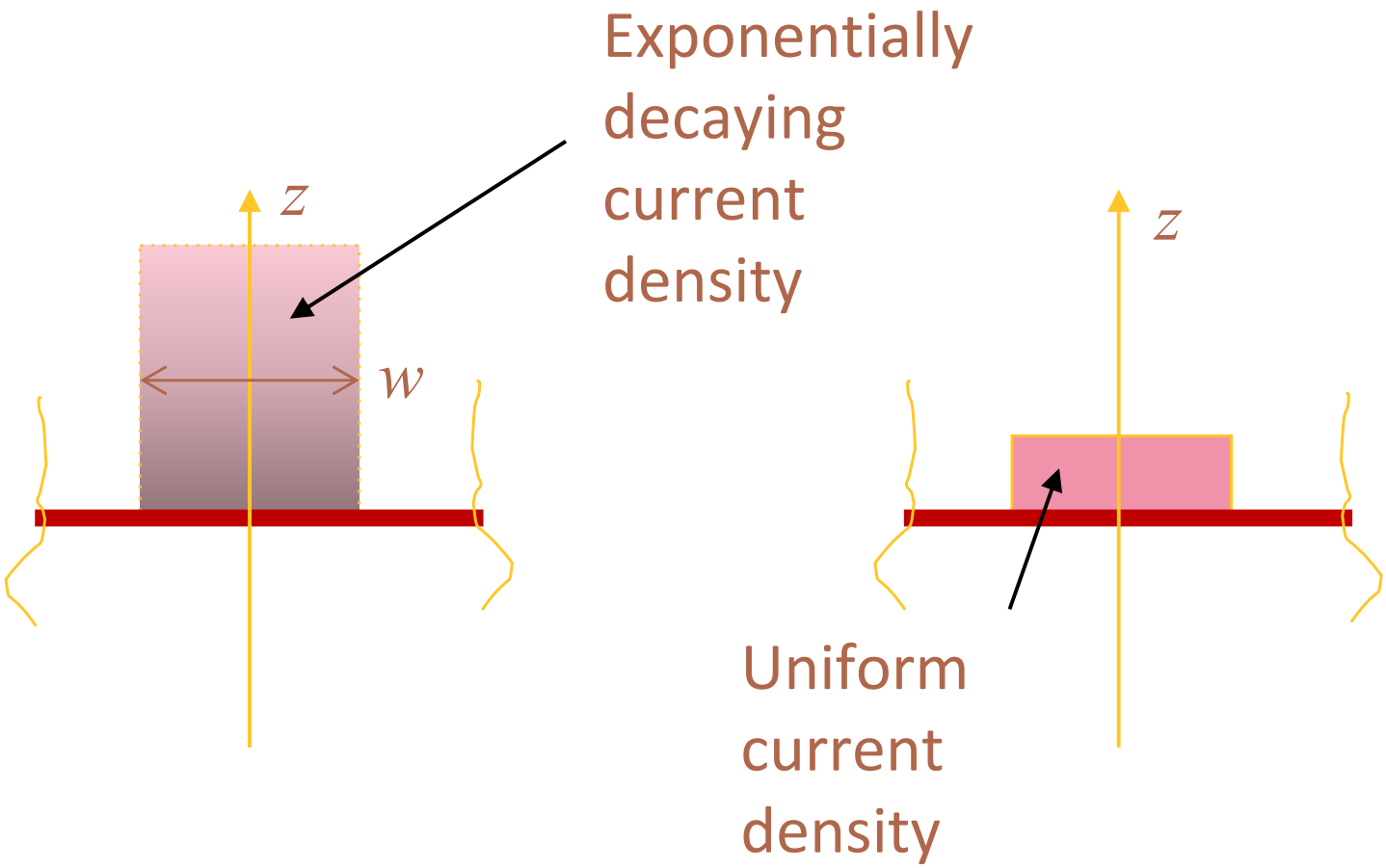
- Notice that the total current is the same as if it resulted from a uniform current density given by

$$\frac{\sigma E_0}{1+j}$$

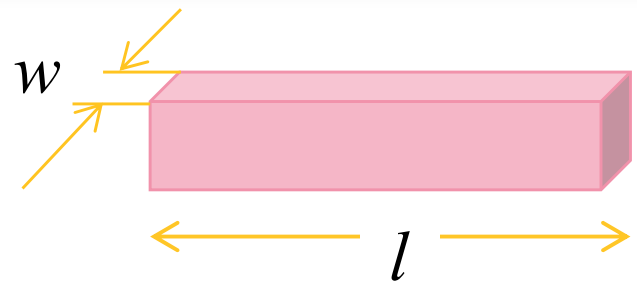
- Flowing over a cross-sectional area given by

$$w\delta_s$$

Semi-Infinite Conducting Material Continued



Surface Impedance



- The voltage across a length l along the direction of current flow (x-direction) of the conductor is

$$\tilde{V} = E_0 l$$

- The impedance of a slab of conductor is given by

$$Z = \frac{\tilde{V}}{\tilde{I}} = \frac{E_0 l}{\left(\frac{\sigma E_0}{1+j}\right) w \delta_s}$$

$$= \left(\frac{1+j}{\sigma \delta_s}\right)\left(\frac{l}{w}\right)$$

Surface Impedance Continued

- We can write

Surface impedance
a.k.a. internal impedance

$$Z = Z_s\left(\frac{l}{w}\right)$$

- Where

$$Z_s = \frac{1+j}{\sigma\delta_s} = R_s + j\omega L_s$$

Surface Impedance Continued

- We have

$$R_s = \frac{1}{\sigma \delta_s} = \sqrt{\frac{\pi f \mu_0}{\sigma}}$$

$$L_s = \frac{1}{\omega \sigma \delta_s} = \frac{1}{2}\sqrt{\frac{\mu_0}{\pi f \sigma}}$$

AC Resistance

- The a.c. resistance of the slab of width w and length l is given by

$$R = R_s\left(\frac{l}{w}\right)$$

- The a.c. resistance per unit length is given by

$$R' = \frac{R_s}{w}$$

The ac resistance per unit length is the surface resistance divided by the cross-sectional width over which the current flows.

Resistance Per Unit Length of Coaxial Cable

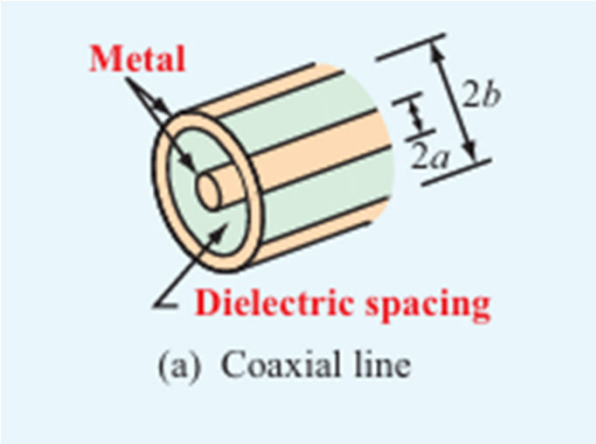
- Consider a coaxial cable with radius of inner conductor given by a and radius of outer conductor given by b.

- The a.c. resistance per unit length of the inner conductor is given by

$$R_1{}' = \frac{R_s}{2\pi a}$$

- The a.c. resistance per unit length of the outer conductor is given by

$$R_2{}' = \frac{R_s}{2\pi b}$$



(a) Coaxial line

Resistance Per Unit Length of Coaxial Cable Continued

- The total resistance per unit length of coaxial cable is

$$R' = R_1{}' + R_2{}' = \frac{R_s}{2\pi}\left(\frac{1}{a} + \frac{1}{b}\right)$$

Question

- What is the equation for the a.c. resistance per unit length for a parallel strips transmission line where the strips have width w?

Summary

- For d.c., the current flows uniformly through the cross-section of the conductor.
- For a.c., we can model the current flow as being uniform through a cross-section as deep as the skin depth.
- The a.c. resistance depends on the surface impedance of the conductor which is a property of the material and the frequency, not the geometry of the conductor.

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